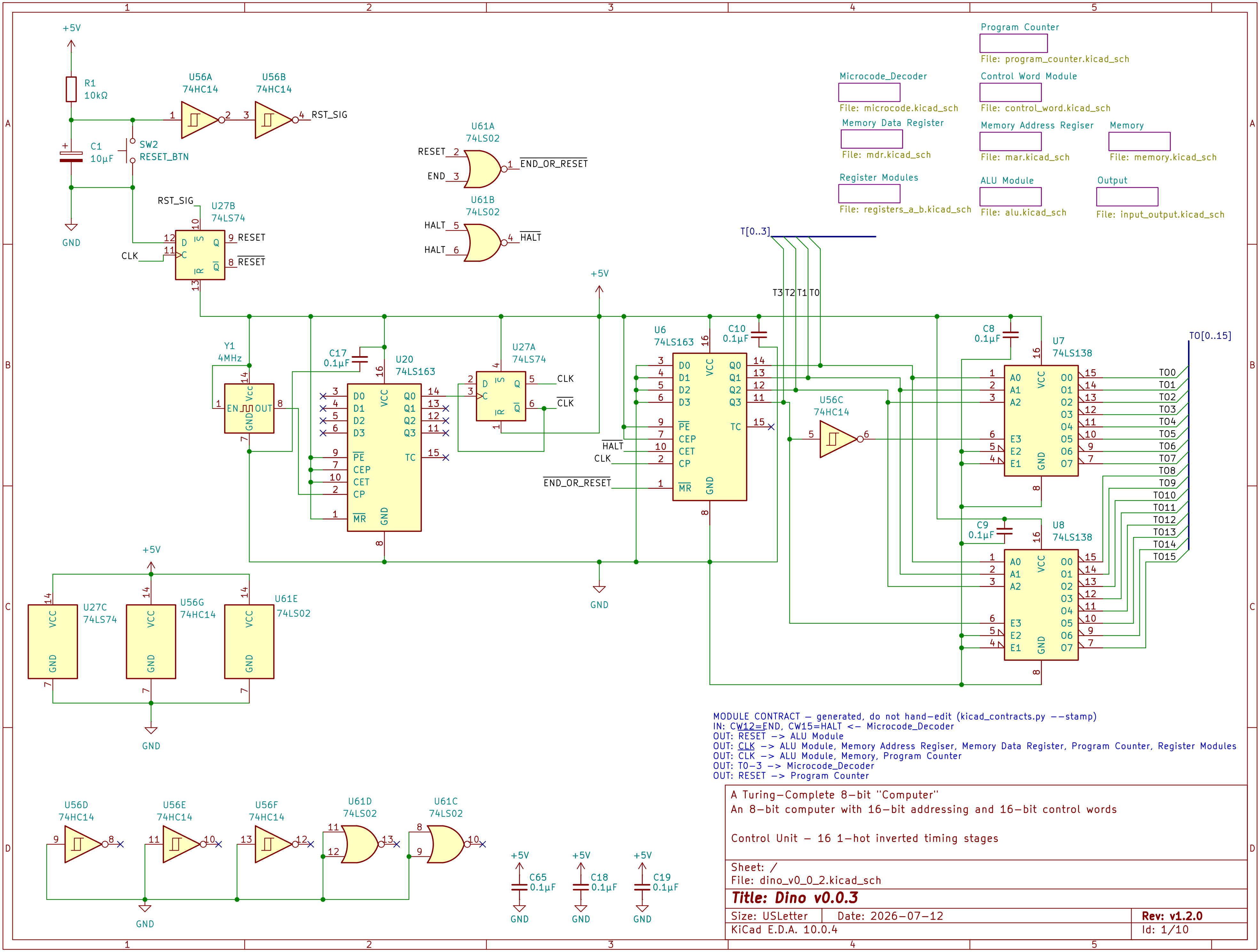


MODULE CONTRACT — generated, do not hand-edit (kicad_contracts.py --stamp)
IN: CW12=END, CW15=HALT <- Microcode_Decoder
OUT: RESET -> ALU Module
OUT: CLK -> ALU Module, Memory Address Regiser, Memory Data Register, Program Counter, Register Modules
OUT: CLK -> ALU Module, Memory, Program Counter
OUT: T0-3 -> Microcode_Decoder
OUT: RESET -> Program Counter

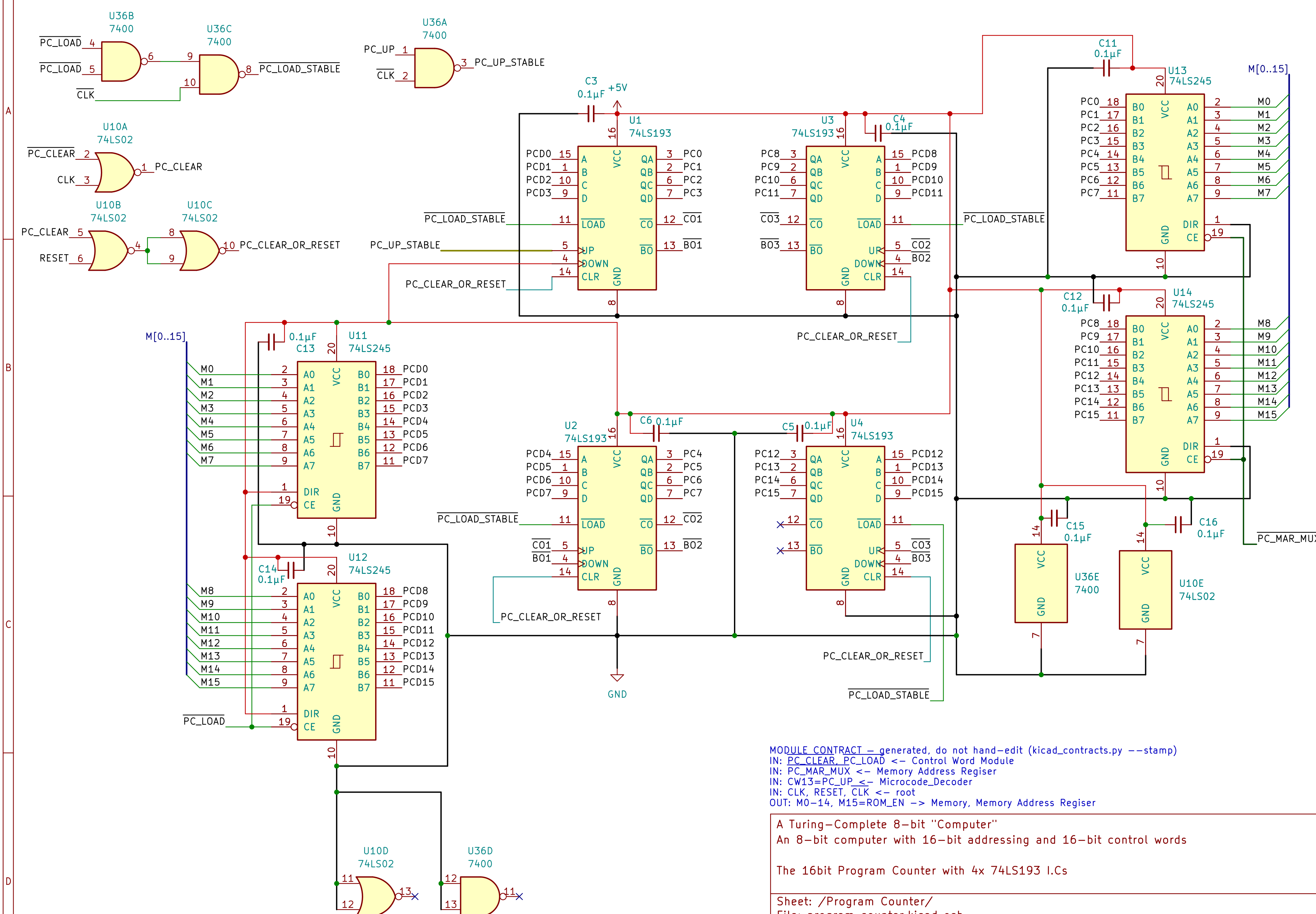
A Turing-Complete 8-bit "Computer"
An 8-bit computer with 16-bit addressing and 16-bit control words

Control Unit - 16 1-hot inverted timing stages

Sheet: /
File: dino_v0_0_2.kicad_sch

Title: Dino v0.0.3

Size: USLetter	Date: 2026-07-12	Rev: v1.2.0
KiCad E.D.A. 10.0.4		Id: 1/10



MODULE CONTRACT — generated, do not hand-edit (kicad_contracts.py --stamp)
IN: PC_CLEAR, PC_LOAD <- Control Word Module
IN: PC_MAR_MUX <- Memory Address Regiser
IN: CW13=PC_UP <- Microcode_Decoder
IN: CLK, RESET, CLK <- root
OUT: M0-14, M15=ROM_EN -> Memory, Memory Address Regiser

A Turing-Complete 8-bit "Computer"
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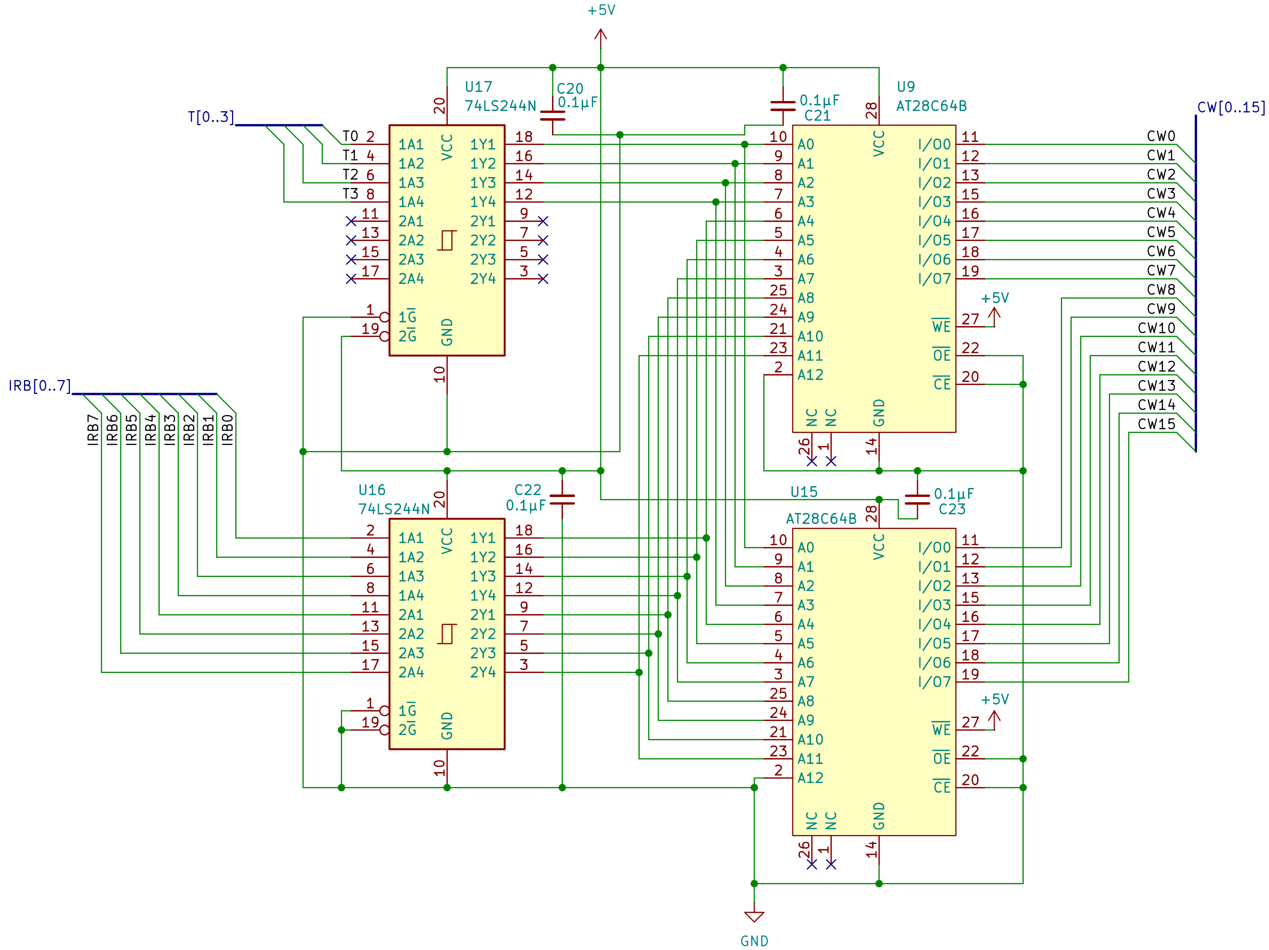
The 16bit Program Counter with 4x 74LS193 I.Cs

Sheet: /Program Counter/
File: program_counter.kicad_sch

Title: Dino v0.0.3

Size: USLetter | Date: 2026-07-12 | **Rev: v1.2.0**
KiCad E.D.A. 10.0.4 | Id: 2/10



MODULE CONTRACT — generated, do not hand-edit (kicad_contracts.py --stamp)
IN: IRB0-7 <- Memory Data Register
IN: T0-3 <- root
OUT: CW10=SA1, CW11=SA0, CW9=SA2 -> ALU Module
OUT: CW0-8 -> Control Word Module
OUT: CW14=PC_MAR_MUX -> Memory Address Regiser
OUT: CW13=PC_UP -> Program Counter
OUT: CW12=END, CW15=HALT -> root

A Turing-Complete 8-bit "Computer"
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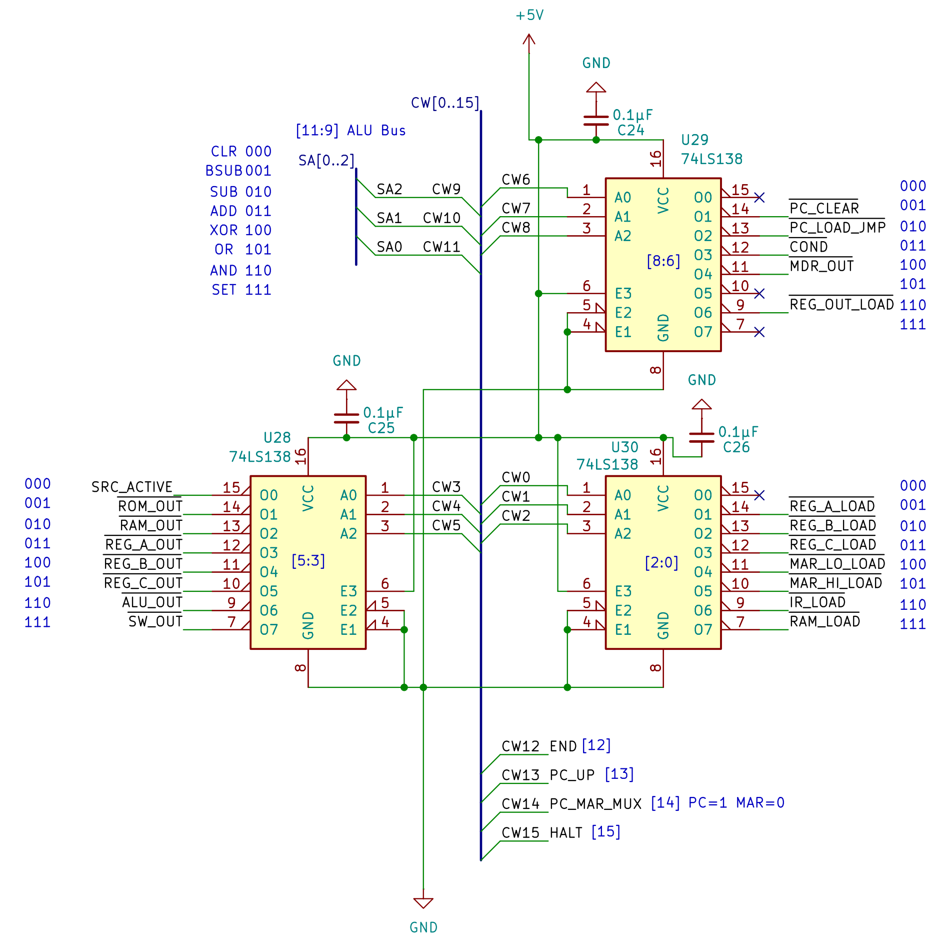
The Microcode Decoder Module

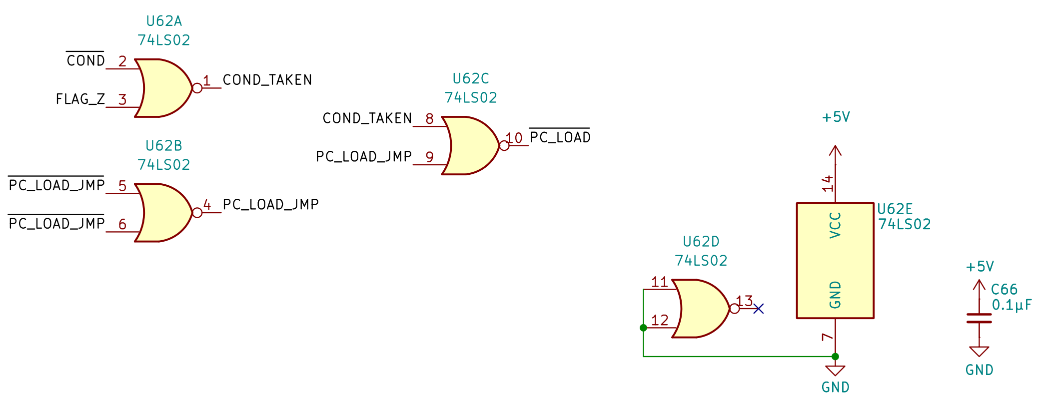
Sheet: /Microcode_Decoder/
File: microcode.kicad_sch

Title: Dino v0.0.3

Size: USLetter | Date: 2026-07-12 | **Rev: v1.2.0**

KiCad E.D.A. 10.0.4 | Id: 3/10





MODULE CONTRACT — generated, do not hand-edit (kicad_contracts.py --stamp)
IN: FLAG_Z <- ALU Module
IN: CW0-8 <- Microcode_Decoder
OUT: ALU_OUT -> ALU Module, Memory Data Register
OUT: REG_A_LOAD, REG_B_LOAD -> ALU Module, Register Modules
OUT: MAR_HI_LOAD, MAR_LO_LOAD -> Memory Address Regiser
OUT: SRC_ACTIVE, IR_LOAD, MDR_OUT, RAM_LOAD -> Memory Data Register
OUT: SW_OUT -> Memory Data Register, Output
OUT: RAM_OUT, ROM_OUT -> Memory, Memory Data Register
OUT: PC_CLEAR, PC_LOAD -> Program Counter
OUT: REG_A_OUT, REG_B_OUT, REG_C_LOAD, REG_C_OUT, REG_OUT_LOAD -> Register Modules

A Turing-Complete 8-bit "Computer"
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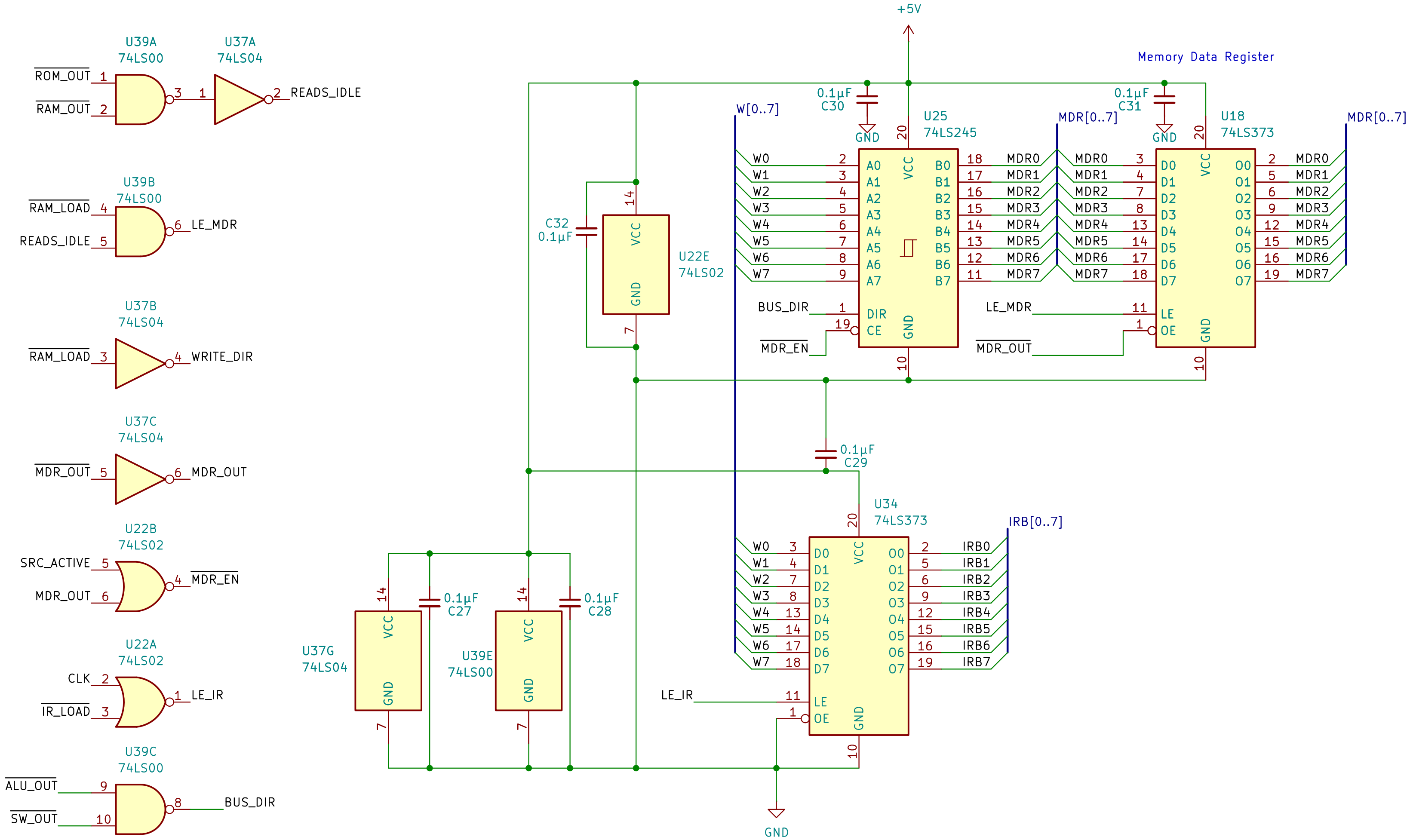
Control Word Module

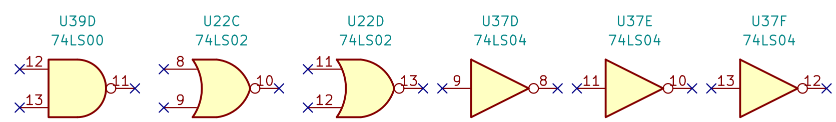
Sheet: /Control Word Module/
File: control_word.kicad_sch

Title: Dino v0.0.3

Size: USLetter	Date: 2026-07-12	**Rev: v1.2.0**
KiCad E.D.A. 10.0.4		Id: 4/10



MODULE CONTRACT — generated, do not hand-edit (kicad_contracts.py --stamp)
IN: SRC_ACTIVE, ALU_OUT, IR_LOAD, MDR_OUT, RAM_LOAD, RAM_OUT, ROM_OUT, SW_OUT <- Control Word Module
IN: CLK <- root
OUT: WRITE_DIR -> Memory
OUT: IRB0-7 -> Microcode_Decoder
BIDIR: W0-7 <-> ALU Module, Memory Address Regiser, Output
BIDIR: MDR0-7 <-> Memory, Register Modules



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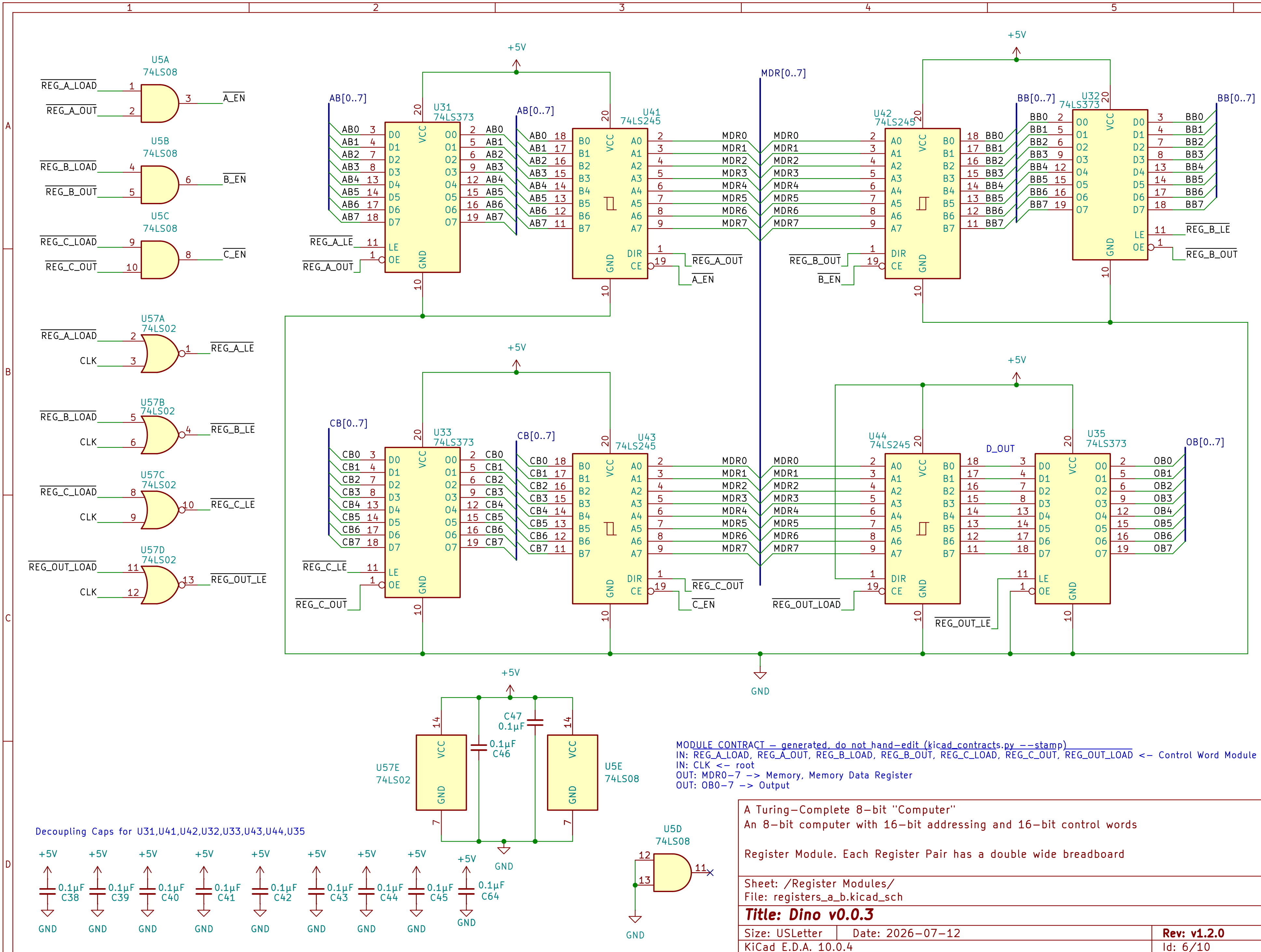
Memory Data Register Module

Sheet: /Memory Data Register/
File: mdr.kicad_sch

Title: Dino v0.0.3

Size: USLetter	Date: 2026-07-12	**Rev: v1.2.0**
KiCad E.D.A. 10.0.4		Id: 5/10



MODULE CONTRACT — generated, do not hand-edit (kicad_contracts.py --stamp)
IN: REG_A_LOAD, REG_A_OUT, REG_B_LOAD, REG_B_OUT, REG_C_LOAD, REG_C_OUT, REG_OUT_LOAD <- Control Word Module
IN: CLK <- root
OUT: MDR0-7 -> Memory, Memory Data Register
OUT: OB0-7 -> Output

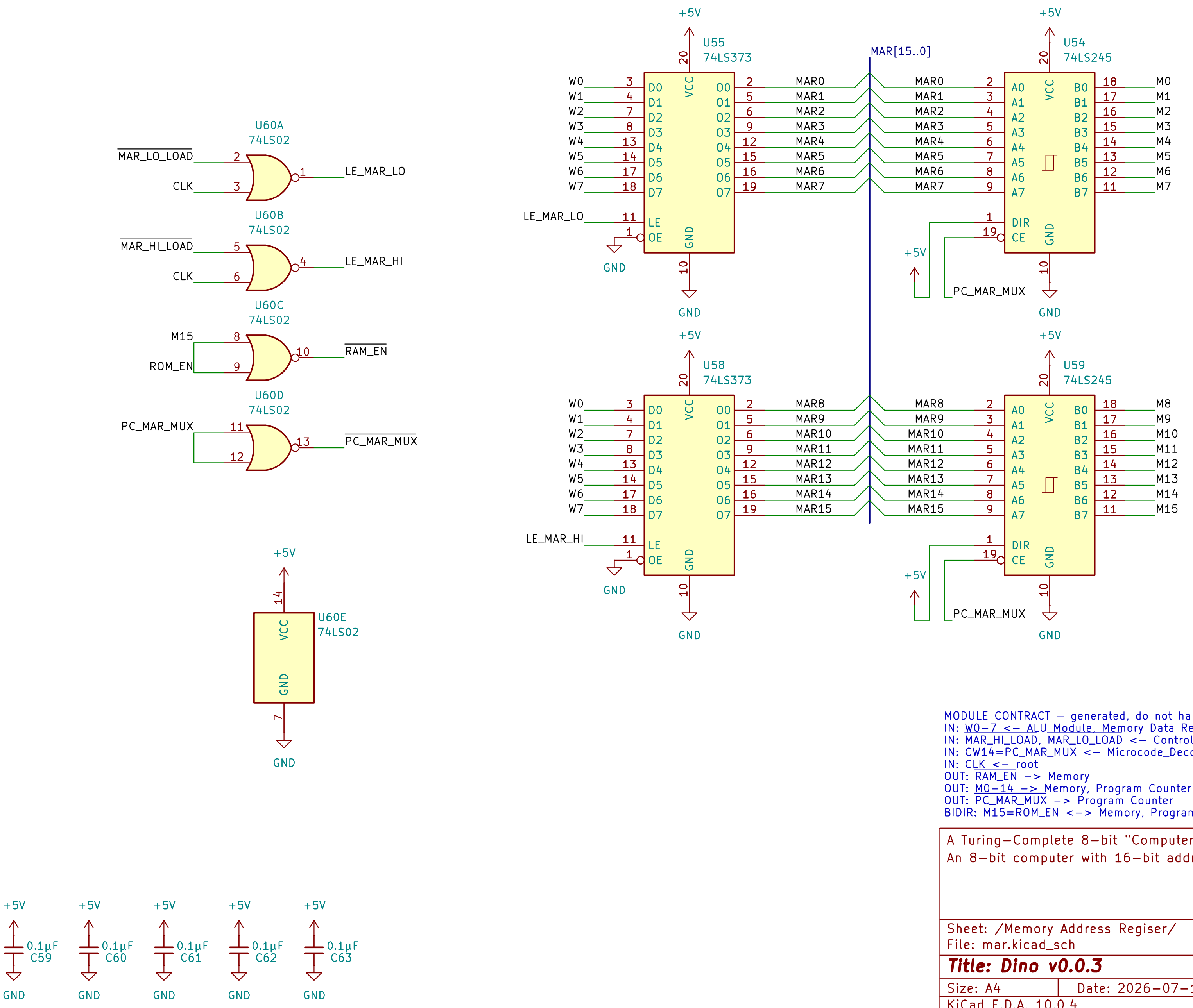
A Turing-Complete 8-bit "Computer"
An 8-bit computer with 16-bit addressing and 16-bit control words

Register Module. Each Register Pair has a double wide breadboard

Sheet: /Register Modules/
File: registers_a_b.kicad_sch

Title: Dino v0.0.3

Size: USLetter	Date: 2026-07-12	**Rev: v1.2.0**
KiCad E.D.A. 10.0.4		Id: 6/10



MODULE CONTRACT – generated, do not hand-edit (kicad_contracts.py --stamp)
IN: W0-7 <- ALU_Module, Memory Data Register, Output
IN: MAR_HI_LOAD, MAR_LO_LOAD <- Control Word Module
IN: CW14=PC_MAR_MUX <- Microcode_Decoder
IN: CLK <- root
OUT: RAM_EN -> Memory
OUT: M0-14 -> Memory, Program Counter
OUT: PC_MAR_MUX -> Program Counter
BIDIR: M15=ROM_EN <-> Memory, Program Counter

A Turing-Complete 8-bit "Computer"
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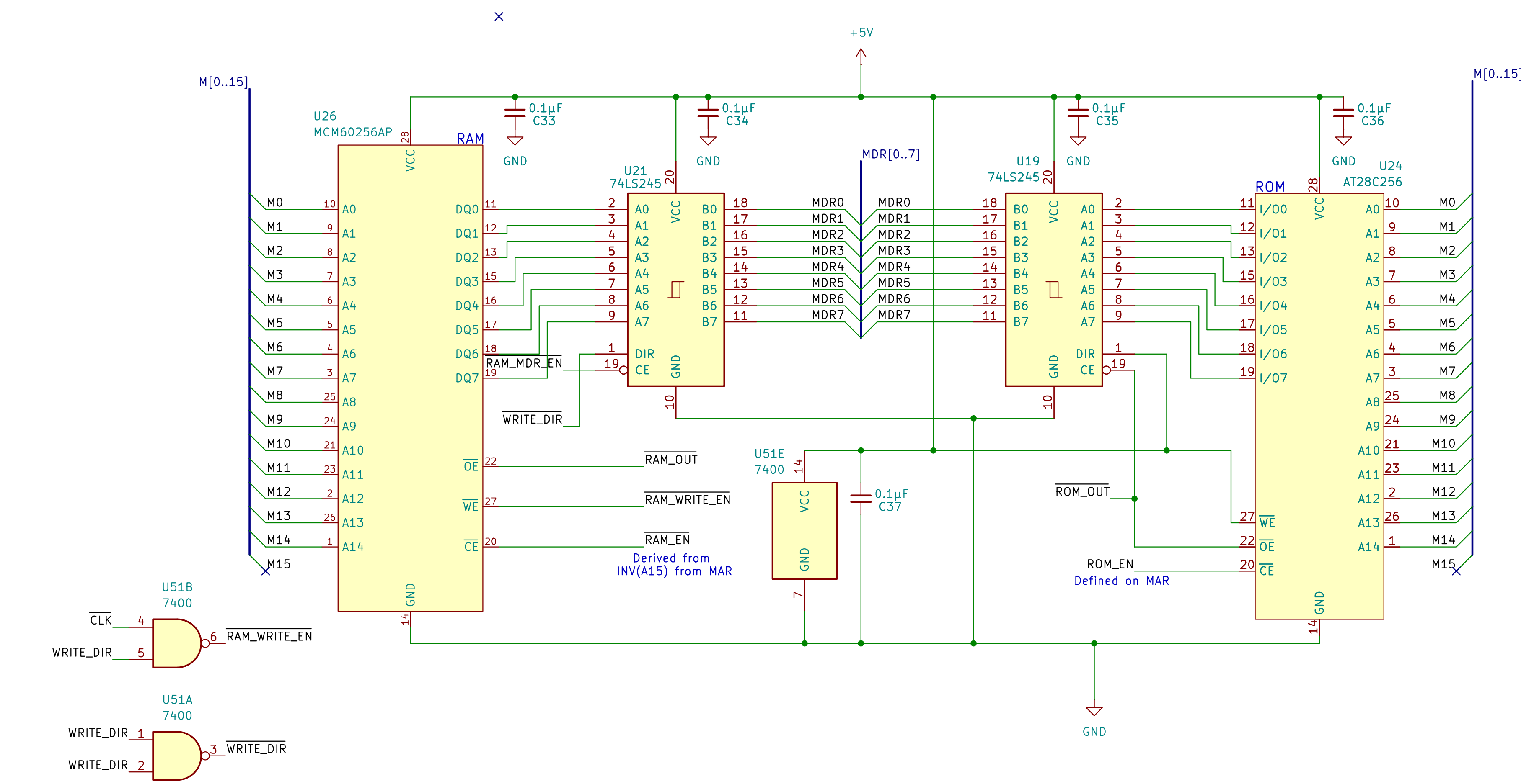
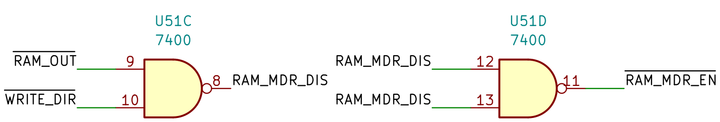
Sheet: /Memory Address Regiser/
File: mar.kicad_sch

Title: Dino v0.0.3

Size: A4	Date: 2026-07-12	**Rev: v1.2.0**
KiCad E.D.A. 10.0.4		Id: 7/10



WRITE_DIR net is defined on MDR



MODULE CONTRACT — generated, do not hand-edit (kicad_contracts.py --stamp)
IN: RAM_OUT, ROM_OUT <- Control Word Module
IN: RAM_EN <- Memory Address Regiser
IN: M0-14, M15=ROM_EN <- Memory Address Regiser, Program Counter
IN: WRITE_DIR <- Memory Data Register
IN: CLK <- root
OUT: MDR0-7 -> Memory Data Register, Register Modules

ROM 0x0000-0x7FFF, RAM 0x8000-0xFFFF, selected by A15

A Turing-Complete 8-bit "Computer"
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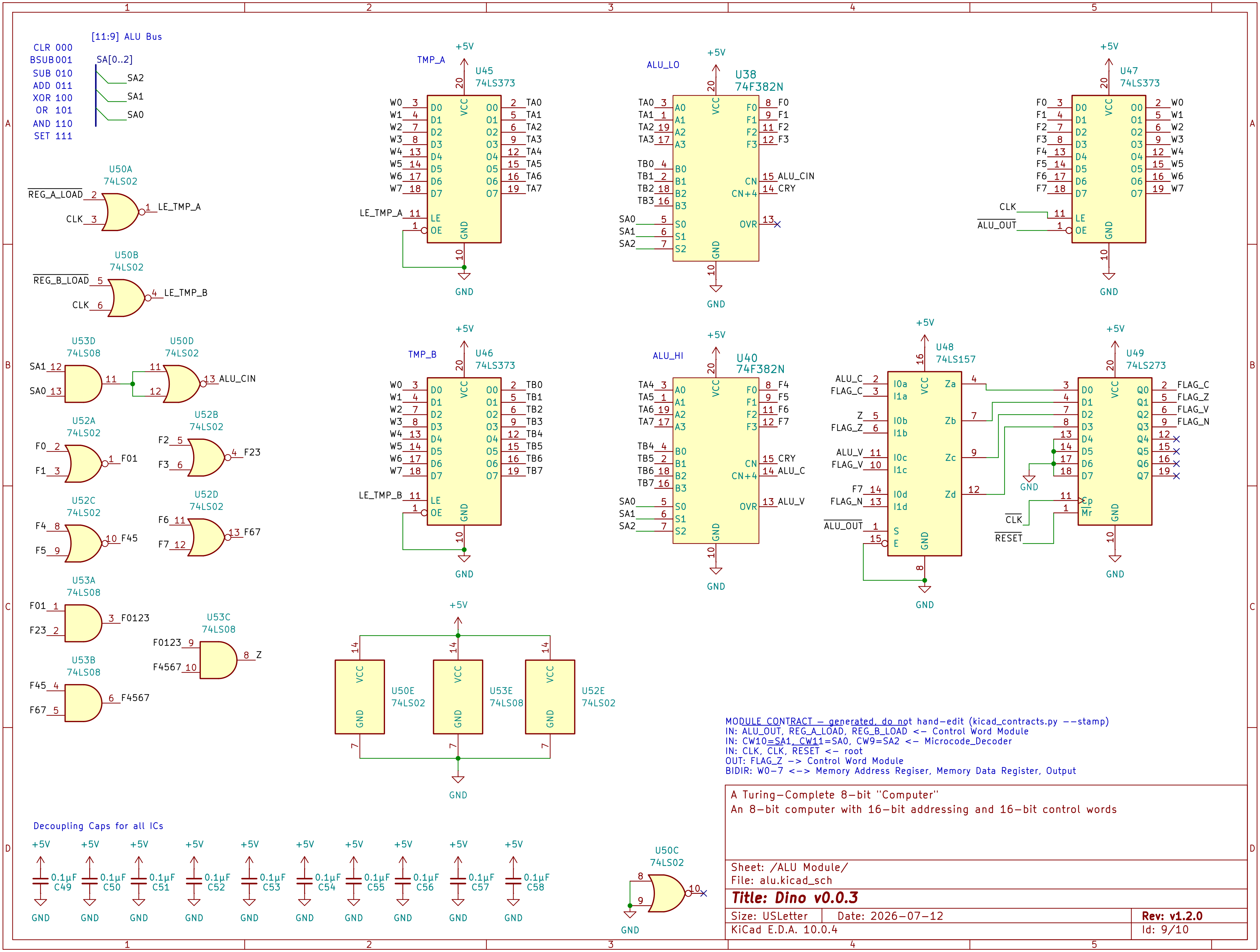
Memory Module

Sheet: /Memory/
File: memory.kicad_sch

Title: Dino v0.0.3

Size: USLetter	Date: 2026-07-12	Rev: v1.2.0
KiCad E.D.A. 10.0.4		Id: 8/10



MODULE CONTRACT — generated, do not hand-edit (kicad_contracts.py --stamp)
IN: ALU_OUT, REG_A_LOAD, REG_B_LOAD <- Control Word Module
IN: CW10=SA1, CW11=SA0, CW9=SA2 <- Microcode_Decoder
IN: CLK, CLK, RESET <- root
OUT: FLAG_Z -> Control Word Module
BIDIR: W0-7 <-> Memory Address Regiser, Memory Data Register, Output

A Turing-Complete 8-bit "Computer"
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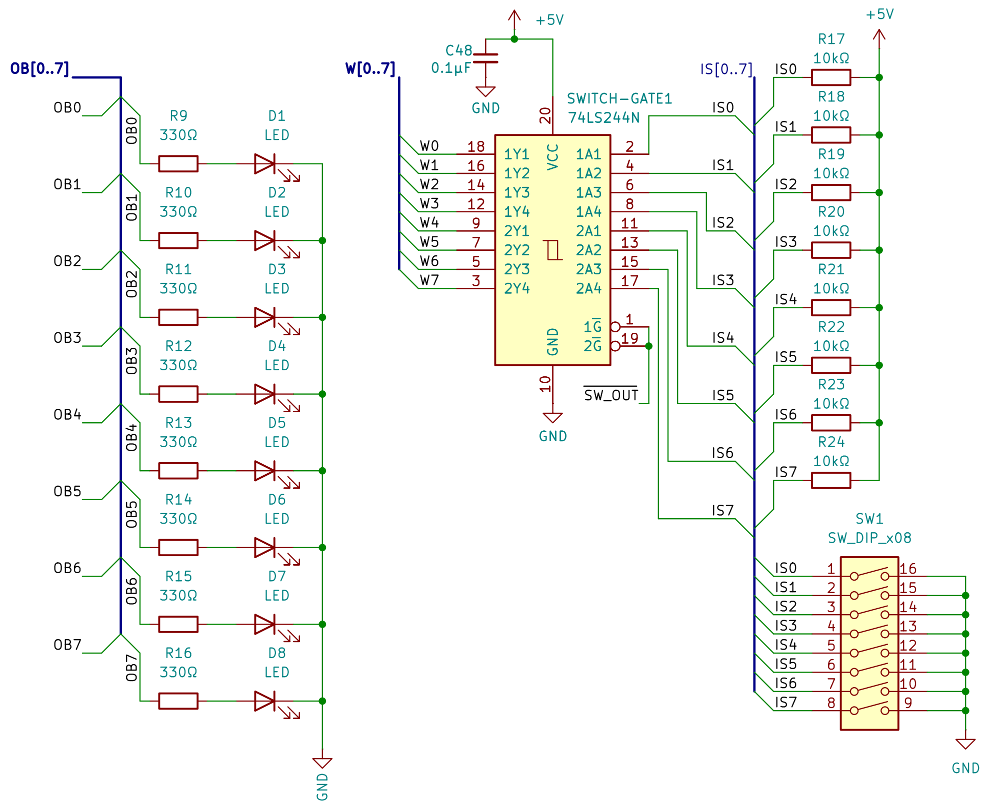
Sheet: /ALU Module/
File: alu.kicad_sch

Title: Dino v0.0.3

Size: USLetter | Date: 2026-07-12 | **Rev: v1.2.0**

KiCad E.D.A. 10.0.4 | Id: 9/10



MODULE CONTRACT — generated, do not hand-edit (kicad_contracts.py --stamp)
IN: SW_OUT <- Control Word Module
IN: OB0-7 <- Register Modules
OUT: W0-7 -> ALU Module, Memory Address Regiser, Memory Data Register

A Turing-Complete 8-bit "Computer"
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Input Output Components

Sheet: /Output/
File: input_output.kicad_sch

Title: Dino v0.0.3

Size: USLetter | Date: 2026-07-12 | **Rev: v1.2.0**
KiCad E.D.A. 10.0.4 | Id: 10/10